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<tr><th colspan="4" rowspan="2">Question</th><th rowspan="2">Marking details</th><th colspan="6">Marks available</th></tr>
<tr><th>AO1</th><th>AO2</th><th>AO3</th><th>Total</th><th>Maths</th><th>Prac</th></tr>
<tr><td></td><td>(b)</td><td></td><td></td><td>Indicative content<br><br>IR spectrum<br>link C—H to 3000 and C—Cl to 650<br><br>Mass spectrum<br>$M_r$ is 126 / 128 / 130<br>two chlorine atoms present from peaks at 35 and 37 / peak height ratios / from $M_r$ differences<br>comment on any fragment<br><br>$^{13}$C NMR<br>two different carbon environments<br>comment on $\delta$ values<br><br>$^{1}$H NMR<br>two different hydrogen environments<br>same number of hydrogens in each environment<br><br>$\Rightarrow$ 1,4-dichlorobutane rather than 2,3-dichlorobutane</td><td></td><td>2</td><td>4</td><td>6</td><td></td><td></td></tr>
<tr><td></td><td></td><td></td><td></td><td colspan="7">5-6 marks<br>At least one valid comment on all four spectra and a correct conclusion for the identity of Y (1,4-dichlorobutane)<br>The candidate constructs a relevant, coherent and logically structured account including key elements of the indicative content. A sustained and substantiated line of reasoning is evident and scientific conventions and vocabulary are used accurately throughout.<br>3-4 marks<br>At least one valid comment on three spectra and good attempt at the identity of Y.<br>The candidate constructs a coherent account including many of the key elements of the indicative content. Some reasoning is evident in the linking of key points and use of scientific conventions and vocabulary is generally sound.<br>1-2 marks<br>At least one valid comment on two spectra .<br>The candidate attempts to link relevant points from the indicative content. Coherence is limited by omission and/or inclusion of irrelevant material. There is some evidence of appropriate use of scientific conventions and vocabulary.<br>0 marks<br>The candidate does not make any attempt or give an answer worthy of credit.</td></tr>
<tr><td></td><td></td><td></td><td></td><td>Question 8 total</td><td>2</td><td>6</td><td>8</td><td>16</td><td>1</td><td>0</td></tr>
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